Symptom-to-Diagnosis Matcher

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Data Structures and Algorithms -3 (25CS2103E)

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TECHNOLOGY/ARTIFICIAL INTELLIGENCE & DATA SCIENCE



DECLARATION

The Mini Project Report entitled "**Symptom-to-Diagnosis Matcher"** is a record of Bonafide work of Rishitha Reddy – 2520090039, Kiran -2520090035, Sahithya – 2520090232**,** submitted in partial fulfillment for the award of B. Tech in Computer Science and Engineering/Computer Science and Information Technology/ Artifical Intelligence & data Science to the K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

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CERTIFICATE

This is to certify that the mini project based report entitled "**Symptom-to-Diagnosis Matcher**" is a bonafide work done and submitted by Rishitha Reddy – 2520090039, Kiran -2520090035, Sahithya – 2520090232, in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** in Department of Computer Science Engineering/ Computer Science and Information Technology/Artificial Intelligence & Data Science, K L (Deemed to be University), during the academic year **2026-2027.**

Signature of the Supervisor

Signature of the HOD **Signature of the External Examiner**

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1.ABSTRACT

Symptom-to-Diagnosis Matcher is a Java-based decision-support engine that maps free-text patient symptoms to likely medical conditions using a structured medical knowledge base. The system first normalizes the entered symptom descriptions and converts them into standard symptom terms. It then compares the identified symptoms with symptoms associated with different candidate diagnoses and assigns transparent scores based on the strength of each match. The candidate diagnoses are ranked according to their scores to identify the most relevant possibilities. The system also recommends a minimal set of diagnostic tests that can best distinguish between the top-ranked conditions. The project demonstrates the application of data structures and algorithms such as hashing, searching, sorting, scoring, ranking, and set based selection. The system is intended as an educational decision-support prototype and does not replace professional medical diagnosis.

2. INTRODUCTION

Medical symptoms can be described in different ways using natural language. For example, users may enter terms such as "high temperature," "feeling hot," or "fever," even though they may refer to the same symptom. This makes direct matching with a medical knowledge base difficult. The Symptom-to-Diagnosis Matcher addresses this problem by normalizing free-text symptoms and matching them with a structured medical knowledge base. The system calculates a transparent score for each candidate diagnosis based on matching symptoms and ranks the possible conditions. The system further analyzes the top-ranked diagnoses and recommends a minimal set of diagnostic tests that can help distinguish between them. Java data structures such as HashMap, HashSet, ArrayList, and PriorityQueue, along with searching, sorting, and scoring algorithms, can be used to efficiently implement the system. The project demonstrates how Data Structures and Algorithms can be applied to build an explainable decision-support system.

3.LITERATURE SURVEY

1. Isabel Symptom Checker: Helps users identify possible medical conditions based on their symptoms. Our project uses a similar symptom-matching approach.

2. Ada Health: Assesses symptoms and provides possible health explanations. Our project uses Java-based symptom matching and ranking.

3. NHS 111 Online: Evaluates symptoms and guides users toward appropriate healthcare services.

4. Mayo Clinic: Provides information about symptoms and their possible causes. Our project compares symptoms with a structured knowledge base.

5. WebMD: Offers symptom-checking resources and medical information about health conditions.

6. Symptomate: Assesses entered symptoms and presents possible health explanations. Our project applies similar concepts using data structures and algorithms.

4.Problem Statement and Objectives

Problem Statement: To develop a Java-based system that normalizes patient symptoms, matches them with possible medical conditions, ranks diagnoses using transparent scoring, and recommends minimal diagnostic tests to distinguish between the top conditions.

Problem Description: The challenge is to develop a Java-based system that matches user-entered symptoms with possible medical conditions and ranks them based on matching scores. It also aims to suggest relevant diagnostic tests to help distinguish between conditions.

Objectives:

- Normalize free-text symptoms into standard terms.
- Matchsymptoms with the medical knowledge base.
- Scorecandidate diagnoses based on symptom matches.
- Rankthe most relevant diagnoses.
- Recommendminimal tests to distinguish between conditions
- Apply DSA conceptssuch as hashing, searching, sorting, and ranking

5.HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements:

- Processor: Intel Core i3 or equivalent and above
- RAM: Minimum 4 GB
- Storage: Minimum 500 MB free space
- Display: Standard monitor
- Input: Keyboard and mouse
- Internet: Required for downloading development tools and optional knowledge-base resources

Software Requirements:

- Operating System: Windows
- Programming Language: Java
- Java Version: JDK 17 or above
- IDE:Visual Studio Code
- Database / Knowledge Base: MySQL
- Frontend: Java Swing • Version Control: GitHub

6.IMPLEMENTATION

Module 1: Symptom Input and Normalization

Code:

```
import java.util.*;
class SymptomProcessor {
    public static Set<String> normalize(String input) {
        Set<String> symptoms = new HashSet<>();
        for (String s : input.toLowerCase().split(",")) {
            symptoms.add(s.trim());
        }
        return symptoms;
    }
}
```

Explanation: Accepts symptoms, converts them to lowercase, and removes duplicates using HashSet.

Module 2: Symptom Matching and Ranking

Code:

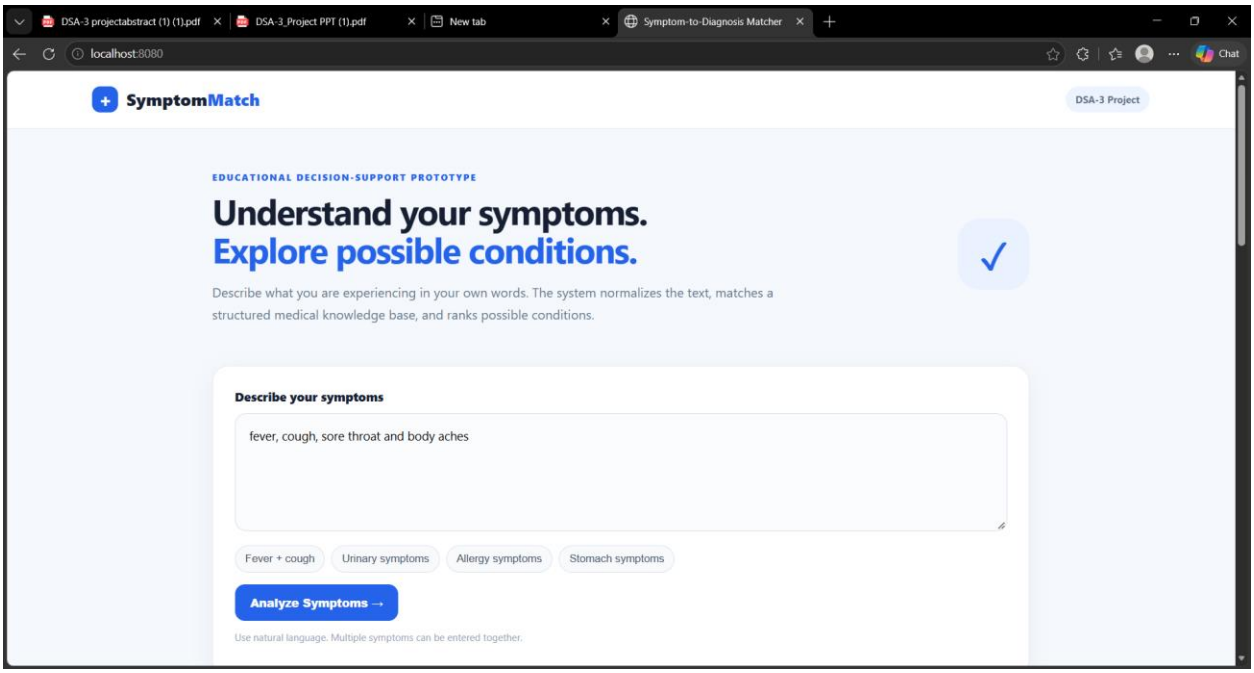
```
static int calculateScore(Set<String> input,
                          Set<String> disease) {
    Set<String> matched = new HashSet<>(input);
    matched.retainAll(disease);

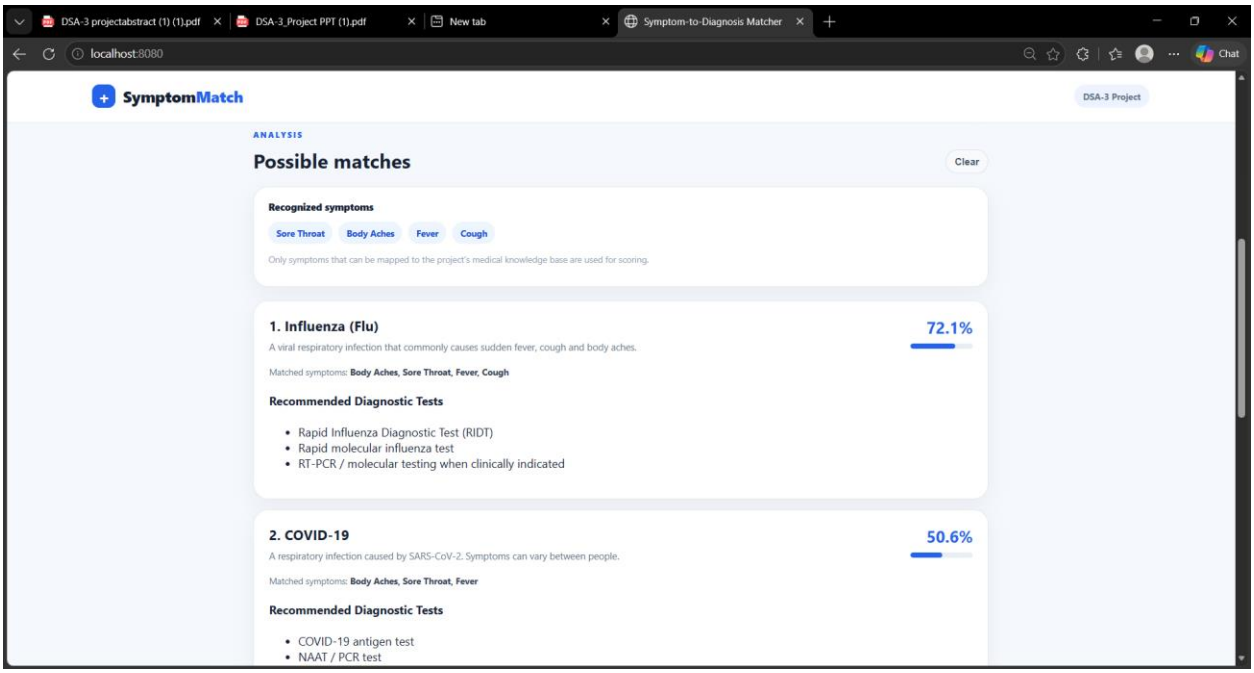
    if (disease.isEmpty()) return 0;

    return matched.size() * 100 / disease.size();
}
```

Explanation: Matches user symptoms with disease symptoms and calculates a matching score to help rank possible conditions.

7.RESULTS & DISCUSSION





8. CONCLUSION & FUTURE SCOPE

Conclusion: The project demonstrates how Java and Data Structures and Algorithms can be used to normalize symptoms, match possible diseases, rank candidates, and recommend useful diagnostic tests.

Future Scope:

Expand the medical knowledge base.

Improve symptom synonym handling.

Consider symptom duration and severity.

Improve diagnostic test selection.

Develop web and mobile versions.

Add multilingual support.

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9.REFERENCES

1. Isabel Symptom Checker — https://symptomchecker.isabelhealthcare.com/
2. Ada Health — https://ada.com/app/
3. NHS 111 Online — https://111.nhs.uk/
4. Mayo Clinic Symptom Checker — https://www.mayoclinic.org/symptom-checker
5. MedlinePlus — https://medlineplus.gov/
6. Cleveland Clinic Health Library — https://my.clevelandclinic.org/health
7. Java Collections Framework — https://docs.oracle.com/en/java/javase/17/docs/api/java.base/java/util/doc-files/coll-overview.html
8. Java HashSet — https://docs.oracle.com/en/java/javase/17/docs/api/java.base/java/util/HashSet.html
9. Java PriorityQueue — https://docs.oracle.com/en/java/javase/17/docs/api/java.base/java/util/PriorityQueue.html
10. Java Collections — https://docs.oracle.com/en/java/javase/17/docs/api/java.base/java/util/Collections.html